

FIT3182 Big Data Management and Processing

Semester 1, 2026

Assignment 3 Specification

Advanced streaming and/or visualisation

Field	Details	Field	Details
Unit	FIT3182: Big Data Management and Processing	Assignment Weight	10%
Assessment Components	Research and code implementation to solve a complex big data stream problem. Presentation of the method through video recording.	Submission Mode	Group (same team formation in Assignment 2)
Release Date	18/May/2026	Due Date	12/June/2026 11:55 PM (Friday, Week 14)
Submission Format	Source code, slides, video recording, readme, confi files.	Primary Focus	Feature deployment innovation Stream join Scalability Research-supported justification

Important note

- This assignment is issued as a specification only. No starter package, starter code, notebook scaffold, or partially completed solution files will be provided.
- Students are expected to independently design the project structure, implementation approach, and presentation.
- The final submission must be reproducible by the teaching team based on the submitted README, deployment instructions, source files, presentation and video recording.

Teaching Team:

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1. Overview

This assignment extends the streaming concepts and implementation foundation from Assignment 2 into a research-focused assessment or a specific innovative extension. You are required to design, implement, deploy, evaluate, and justify a scalable stream analytics solution in a cloud or cloud-simulated environment.

The primary objective is to move beyond prototype-level development and demonstrate a more production-oriented workflow, innovation, or advanced methods that encompass deployment architecture, operational reasoning, stream join design, scalability evaluation, and research-supported technical justification.

2. Learning Outcomes

- a) Apply modern big data processing tools for various data structures
- b) Solve complex big data streaming problems

3. Assignment Theme

Students (*working in the same team formation from Assignment 2*) are required to extend the streaming application from Assignment 2 into one of the following:

- a) Cloud-deployable Spark and Kafka using, for instance, an AWS Free Tier Account.
- b) Using Spark Machine Learning (or Spark ML) library to analyse the streams and make predictions.
- c) Modify the stream join algorithm using join algorithms published in the research literature.
- d) Implementing the stream processing and join using Kafka streams (no Spark)
- e) Interactive visualisation (e.g., dashboard illustrating impactful information from Assignment 2).
- f) Any additional innovation approved by the teaching staff.

Please pick one of the above.

The final solution should explicitly demonstrate the following five dimensions:

- a) Selected theme extending assignment 2
- b) Deployment and reproducibility
- c) Innovation and technical complexity
- d) Performance analysis
- e) Documentation, presentation, communication

Important - In this assignment, you and your team partner will use your training and knowledge from this unit to conduct research to identify methods or techniques to solve a particular problem based on the aforementioned theme(s). Therefore, you should research, learn and apply new tools or methods (where applicable) in this assignment. For instance, if students intend to use the Spark ML library to analyse streams and make predictions, they should learn on their own how to use it. The same applies to the rest of the themes. Through research, you will learn to discover solutions. Therefore, please use this assignment as an opportunity to learn to become an active problem-solver and build the critical thinking skills required to navigate complex problems. You can also use generative Artificial Intelligence (AI) to aid in this assignment, as outlined in Section 9 below.

4. Problem Statement

Assume you are part of a data engineering team tasked with operationalising a streaming analytics platform for real-world deployment. A prototype has already demonstrated stream ingestion, joining, sink integration, event detection and visualisation. The next business requirement is to evolve the prototype

within the specified scope (see Section 3 above) into a deployable solution with research-supported technical decisions.

Your solution should therefore demonstrate correct implementation, with architectural or algorithmic justification, analysis, and a critical discussion of trade-offs.

5. Technical and Execution Requirements

- a) All tasks should be implemented in Python (.py or Jupyter notebook) or a programming language that best implements the solution.
- b) You may use cloud-native or cloud-simulated deployment approaches such as Docker, virtual machines, or managed/self-hosted services, provided the submission remains reproducible.
- c) You should document the implementation, architecture, deployment steps, configuration assumptions, and operational limitations of the algorithm.
- d) You should not submit screenshots or static outputs as a substitute for an actual implementation. All evidence must be generated by the submitted code and deployment workflow.

6. Submission Requirements

Please compress your files into a single zipped file when submitting the work to Moodle. To reiterate, although this is a group assignment, each team member must submit the assignment on Moodle.

The archive should contain the following:

Required submission components

Component	Requirement	Notes
Source files	All source files relevant to the project and implemented methods.	The codebase should be modular and runnable by the teaching team.
Deployment configuration	Include deployment scripts, container files, and configuration templates where applicable	The deployment workflow must be reproducible.
README	Single readme file explaining project structure, dependencies, setup, and how to run the solution.	Commands and assumptions must be clear and concise.
Technical presentation slides	Submit slides describing the selected theme, implementation, results and/or analysis, cited research article(s). The slides should be either in PPTX or PDF format.	The slides should summarise architecture, innovation, scalability findings, and research support.
Video recording of the presentation slide and demonstration of the application	The recording should consist of presenting the slides and demonstrating the running application.	The recording should not exceed 10 minutes. All team members should speak in the recording with the camera turned on. You can use Zoom to record the screen and presentation. If you have another recording application, please feel free to use it. The recording should be in the MP4 file format.

Component	Requirement	Notes
Research article(s)	Research articles referred to in this project (if any)	The complete citation to the research article should be included in the presentation slides.

Recommended project structure

```
<###>_assignment03/
├── README.md
├── slides.pdf
├── source file(s)
├── deployment/
│   ├── scripts/
│   └── config/
├── Video recording
└── research article(S) /
    └── article.pdf
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7. Assessment marking structure

The following table outlines the marking structure for this assignment.

Criterion	Marks	Description
1. Extension of the Assignment 2 pipeline and functional implementation, backed by research articles	2.0	Correctly extends the Assignment 2 pipeline using one selected theme from Section 3. Includes relevant research or technical literature to support the choice of theme or method.
2. Deployment and reproducibility	1.5	Provides runnable code, configuration, setup instructions, and evidence that the solution can be reproduced.
3. Innovation and technical complexity	3.0	Explains what is new or improved compared with Assignment 2 and justifies key design choices. Includes relevant research or technical literature to support technical innovation.
4. Performance analysis	1.5	Provides reasonable evaluation, discussion, or experiment related to scalability, throughput, latency, resource use, or complexity.
5. Documentation, presentation, communication	2.0	Communicates the method and analysis through slides, code/application demonstrations, and a recorded video.

Total	10	
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This assignment will be assessed on the extent to which students extend their Assignment 2 streaming pipeline into a technically meaningful, reproducible and research-supported solution. A Pass grade for this assignment requires a small but working extension of the Assignment 2 pipeline. Credit and Distinction grades require stronger implementation quality, clearer deployment or reproducibility, and meaningful discussion of scalability and trade-offs. High Distinction work must demonstrate substantial innovation, strong technical execution, clear scalability or operational evaluation, and well-integrated justification based on relevant research literature.

There is no interview for this assignment. However, the teaching staff may reach out to you and/or your teammate for further clarification regarding the submitted work. If you do not respond to the teaching staff, it could affect their assignment marks.

8. Academic Integrity and Late Submission

- a) Each team must ensure that the submitted work reflects their own accountable academic effort.
- b) Members of a team can share the same submission content for the assignment. However, exact content sharing between teams is prohibited as it will trigger collusion. Multiple teams can work on the same theme and use the same research article. However, each team should submit their own developed work.
- c) Late penalties, extensions, and special consideration are to be handled in accordance with official University procedures and Moodle guidance.

9. Generative AI Use Declaration

You may use approved generative AI tools as a support aid for research, learning, debugging, documentation, or implementation assistance. You and your teammate remain fully responsible for the correctness, verification, understanding, and accountable authorship of the final submission.

Recommended declaration format in the slides (at the beginning or at the end of the slides): Generative AI Declaration: We used [tool name] for [purpose]. The prompt(s) used were: [insert prompt(s) or concise prompt summary]. We reviewed, tested, and verified the final submitted work.